

SiPM Operating-Point Calibration for the gLOWCOST Detector

Tharindu Hettiarachchi

Department of Physics and Astronomy, Georgia State University

September 2026

Abstract

The gLOWCOST detector uses plastic scintillator tiles, wavelength-shifting fibres, and silicon photomultipliers (SiPMs) to detect charged particles. Stable operation requires an SiPM bias that is high enough to detect scintillation light but not so high that dark counts, correlated avalanches, or readout saturation dominate. This work develops and tests a waveform-based method for measuring the breakdown voltage of Hamamatsu S13360-2050VE SiPMs through the existing gLOWCOST analogue readout. Dark-pulse waveforms were recorded with a PicoScope, corrected event by event for baseline offset, and integrated in a peak-aligned time gate. The spacing between neighboring photoelectron populations was used as a relative gain measurement and extrapolated linearly to zero spacing.

The study also establishes an EPIC-tile coincidence method for measuring the full response of GSU gLOWCOST tiles. This zero-event measurement is intended to test whether channels operated at equal overvoltage have equal practical detection efficiency. The current scintillator data are treated as method development rather than a completed photon-detection-efficiency measurement. Once the method has been validated, a pulsed-LED study can provide a true SiPM zero-event calibration.

Contents

1	Introduction	3
2	Detector and Readout	4
2.1	SiPM	4
2.2	gLOWCOST analogue path	4
2.3	Bias generation	5
2.4	Waveform instrumentation	5
3	Measurement Principle	5
3.1	Breakdown voltage and overvoltage	5
3.2	Why pulse area was used	6
4	Experimental Procedure	6
4.1	Dark-pulse scans	6

4.2	Channel mapping and swaps	7
4.3	Trigger study on PCB CH3	7
5	Waveform Processing	8
5.1	Event-by-event baseline	8
5.2	Pulse location and integration	8
5.3	Saturation and quality control	8
6	Photoelectron Spectrum Analysis	8
6.1	Historical peak-finding method	8
6.2	Pedestal-constrained comb model	9
6.3	Why every visible maximum is not a p.e. peak	9
6.4	Spacing and breakdown-voltage uncertainties	9
7	Breakdown-Voltage Results	11
7.1	Original SiPM pair	11
7.2	Lower-bias limit	11
7.3	SiPM 1 ($\triangle$) and SiPM 2 ($\star$)	12
7.4	Provisional operating points	12
8	EPIC-Triggered Zero-Event Study	14
8.1	Purpose	14
8.2	Zero fraction and dark correction	14
8.3	Pilot observations	15
8.4	Production acquisition	15
8.5	Zero-event results	15
9	Discussion	16
9.1	Temperature	16
10	Next Measurements	16
11	Conclusion	16
	Acknowledgements	16
A	Software and Reproducibility	17
B	Important Distinction Between Thresholds	17
C	Summary of Uncertainty Categories	17

1 Introduction

The gLOWCOST detectors are being developed as compact and inexpensive charged-particle detectors. A particle crossing a plastic scintillator deposits energy and produces optical photons. A wavelength-shifting fibre collects part of this light and transports it to a silicon photomultiplier. The SiPM converts the light into a fast electrical pulse, which is amplified, compared with a threshold, and processed by an FPGA. The processed counts are sent to the Raspberry Pi that hosts the gLOWCOST readout, where the data are prepared and transferred to the Georgia State University servers.

A silicon photomultiplier is a compact light sensor made from many small avalanche-photodiode cells connected in parallel. Each cell is operated above its breakdown voltage and produces a nearly fixed amount of charge when it is triggered. When several cells fire together, their charges add and form a larger pulse. This makes it possible to detect the weak light from a scintillator and, under suitable conditions, separate signals corresponding to different numbers of fired cells. SiPMs are well suited to a particle detector such as gLOWCOST, but their gain, noise, and photon response depend on the applied bias and temperature. Their operating voltage must therefore be measured and controlled carefully [1, 2].

The operating voltage is one of the most important detector settings. An SiPM must be biased above its breakdown voltage, but the breakdown voltage is not identical for every device and changes with temperature. Raising the voltage increases the charge produced by one fired microcell and generally increases photon-detection efficiency. It also increases dark-count rate, optical crosstalk, and afterpulsing [2, 1]. The best setting is therefore a measured operating point, not simply the highest allowed voltage.

This study began with a practical question: can the existing gLOWCOST readout be used to calibrate each SiPM without adding a separate charge-sensitive instrument? The work developed through several stages. Early manual waveform recordings showed photoelectron structure but also exposed problems with baseline treatment, trigger selection, and channel identity. The acquisition was then automated, the pulse-area method was made more robust, complete scans were repeated, and channel swaps were used to distinguish SiPM behavior from readout-path behavior. Once the SiPMs are calibrated and their photon-detection efficiencies are measured, the gLOWCOST detectors can be adjusted to comparable sensitivity, making future cosmic-ray measurements more reliable.

The objectives were:

1. resolve photoelectron populations using the amplified analogue waveform;
2. determine the response corresponding to one additional fired microcell;
3. measure breakdown voltage from the gain-related spacing;
4. test repeatability and sensitivity to trigger conditions;
5. set provisional equal-overvoltage operating points; and
6. prepare an independent particle-trigger measurement of full-chain efficiency.

The last objective is important. Equal overvoltage is a useful starting condition, but it does not prove equal photon sensitivity. The scintillator, fibre, optical coupling, SiPM, amplifier, and software threshold all contribute to the response seen in the final detector.

2 Detector and Readout

2.1 SiPM

The devices studied were Hamamatsu S13360-2050VE MPPCs. The series datasheet gives a $2\text{ mm} \times 2\text{ mm}$ photosensitive area, $50\text{ }\mu\text{m}$ pixel pitch, 1584 pixels, a typical breakdown voltage near 53 V , and a typical gain of approximately 1.7×10^6 under the stated manufacturer conditions [3]. These are typical series values. They do not replace a measurement of the individual device installed in a detector.

An SiPM contains many microcells operated in Geiger mode. One carrier initiating an avalanche produces the charge of one fired cell. More than one fired cell produces a larger pulse. Repeated dark pulses can therefore form approximately equally spaced populations corresponding to one, two, three, or more fired cells.

2.2 gLOWCOST analogue path

The waveform was measured after the gLOWCOST analogue amplifier and before the comparator. The recorded signal therefore includes the SiPM response, the coupling network, the amplifier, cable and probe effects, and the PicoScope response. The comparator and FPGA are used during normal counting, but they were not used to measure the photoelectron spacing.

The non-inverting amplifier has a nominal gain of 101. This large gain makes single-cell structure visible on the scope, but it also means that saturation and bandwidth must be checked. A stable p.e. spacing measures the complete analogue-chain response to one additional fired cell. It is not automatically the absolute SiPM charge in coulombs.

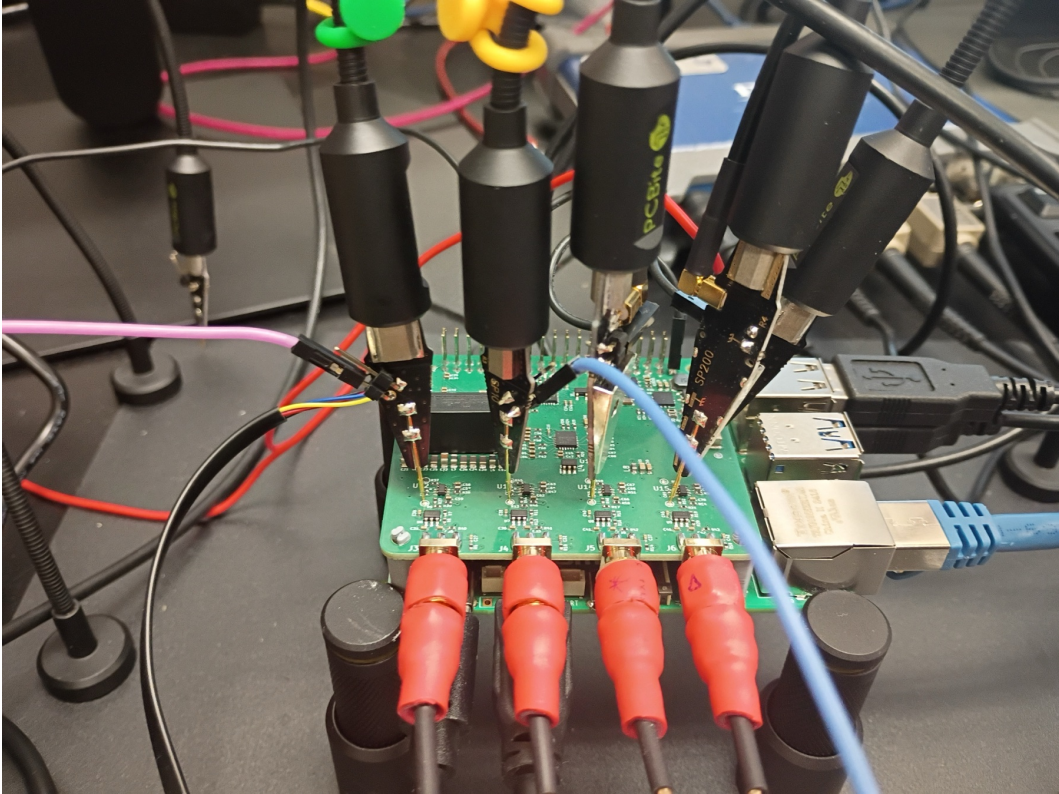


Figure 1: The four-channel gLOWCOST readout connected to the PicoScope. The analogue outputs were measured before the comparator.

2.3 Bias generation

The detector uses a common high-side voltage from a MAX1932 circuit and an independently adjustable low-side voltage from a channel DAC. The effective SiPM bias is

$$V_{\text{bias}} = V_{\text{high}} - V_{\text{low}}. \quad (1)$$

The MAX1932 interface is write-only in the present hardware. A control code can be converted into a predicted voltage using the confirmed calibration relation, but this is not a live measurement at the SiPM terminals. The channel DAC also has a calibrated code-to-voltage relation. Their difference was used for the bias axis in this study.

Temperature compensation was disabled during controlled breakdown-voltage scans so that the voltage did not move while one point was recorded. Temperature was logged and, where stated, bias values were normalized using the provisional manufacturer coefficient of $54 \text{ mV } ^\circ\text{C}^{-1}$.

2.4 Waveform instrumentation

The main measurements used a PicoScope 3406B with 10:1 passive probes. The primary automated scans used a 1 ns sample interval and 10,000 waveforms at each voltage point. The acquisition configuration recorded probe ratio, range, coupling, trigger, time window, channel mapping, temperature, and bias commands.

The 10:1 correction is essential. Applying it twice would make the apparent signal ten times too large; omitting it would make the result ten times too small. The correction was therefore treated as part of the acquisition metadata rather than an informal plotting adjustment.

3 Measurement Principle

3.1 Breakdown voltage and overvoltage

The operating quantity is the overvoltage,

$$V_{\text{OV}} = V_{\text{bias}} - V_{\text{br}}, \quad (2)$$

where V_{br} is the breakdown voltage. Above breakdown, the avalanche charge from one microcell is approximately proportional to overvoltage over a useful operating interval. If the analogue readout remains linear, the spacing between neighboring photoelectron populations follows the same behavior,

$$\Delta Q = m(V_{\text{bias}} - V_{\text{br}}) = mV_{\text{bias}} + b, \quad (3)$$

where ΔQ is the measured pulse-area spacing and m includes the SiPM cell capacitance and the readout conversion. The breakdown voltage is the x-intercept,

$$V_{\text{br}} = -\frac{b}{m}. \quad (4)$$

This does not mean that a visible waveform is measured at zero amplitude below breakdown. The line is fitted to resolved photoelectron spacing above breakdown and extrapolated to the

point where the gain-related spacing becomes zero. This use of gain extrapolation is established in SiPM characterization [4, 2].

3.2 Why pulse area was used

Peak height is intuitive, but it depends on pulse width, sampling phase, ringing, and clipping. The primary observable in the final method was a peak-aligned integral,

$$Q_{\text{proxy}} = \int_{t_{\text{peak}} - 20 \text{ ns}}^{t_{\text{peak}} + 200 \text{ ns}} v(t) dt. \quad (5)$$

Its unit is mV ns. It is proportional to charge after the analogue chain, but it is called a charge proxy because the transfer impedance and full frequency response have not yet been calibrated. Peak height was retained as an independent cross-check.

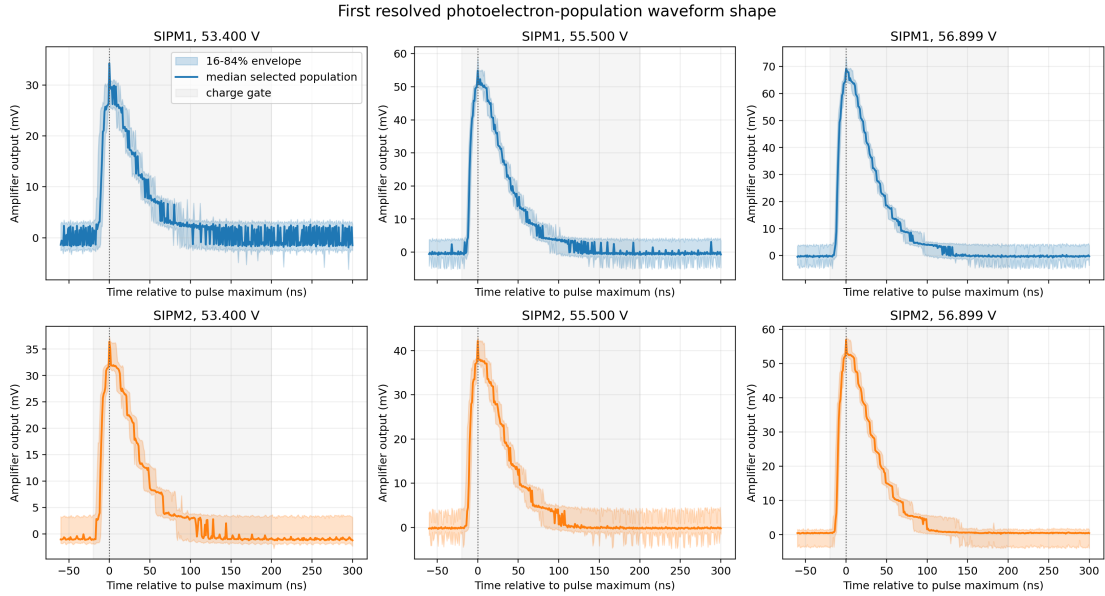


Figure 2: Representative baseline-subtracted single-photoelectron waveform shapes used to define the prompt pulse region.

4 Experimental Procedure

4.1 Dark-pulse scans

The SiPM was light blocked. Thermal carriers still initiate avalanches, and optical crosstalk can make more than one microcell fire. These events provide the p.e. populations without an external light source.

For each bias point, the voltage was set, the hardware was allowed to settle, temperature was recorded, and the requested waveforms were saved. Each SiPM triggered on its own waveform. The automated sequence also recorded the final hardware state and was designed to turn off the high voltage after completion or failure.

An independent HV-off acquisition measured the electronic pedestal. The pedestal is the zero-avalanche response; it is not necessarily the first peak in a self-triggered dark-pulse histogram.

The pedestal location and width were used to constrain the zero-p.e. population in the spectrum model.

4.2 Channel mapping and swaps

Physical SiPM labels, PCB channels, and scope channels were recorded separately. This was necessary because connections were deliberately exchanged during diagnosis. A result was assigned to a physical SiPM only when the mapping was known.

Two physical groups appear in the study:

Table 1: SiPM groups used during the development of the method.

Group	Device labels	Main purpose
Original pair	SIPM1, SIPM2	Method development and repeated scans
Later pair	SiPM 1 (Δ), SiPM 2 ($\star$)	Channel diagnosis and clean repeats

The numerical results from these groups are not combined because they are different physical devices.

4.3 Trigger study on PCB CH3

The first measurements through PCB CH3 had very low acceptance at a 15 mV PicoScope trigger. Swapping scope inputs did not move the behavior; it remained with the PCB path. A dedicated trigger scan at an effective bias of 56.401 V found a sharp transition between 15 mV and 25 mV. Acceptance increased from 1.62% to 99.82%, while the extracted p.e. spacing remained stable from 20 to 45 mV. A 25 mV acquisition trigger was therefore used for the clean SiPM 1 (Δ) and SiPM 2 ($\star$) scans.

This threshold was a PicoScope acquisition setting. It was not the comparator threshold used for normal detector counting, and it did not remove the underlying CH3 feature from the electronics.

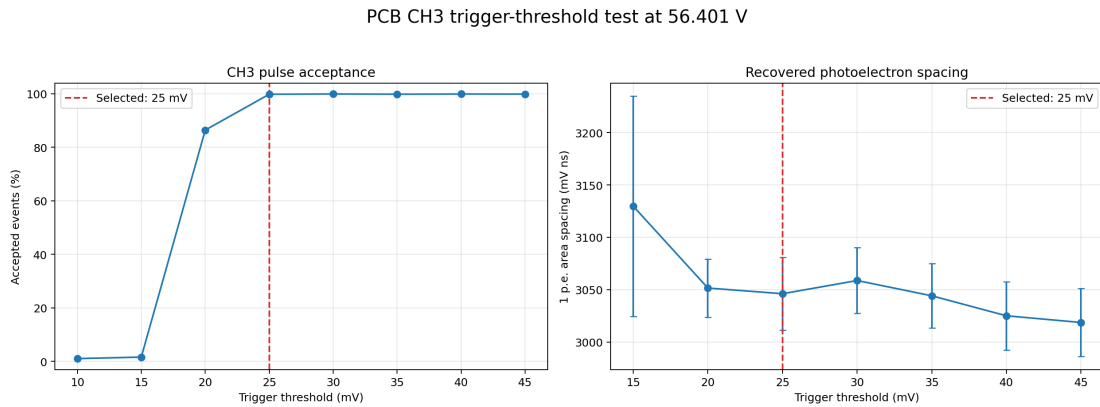


Figure 3: PCB CH3 acquisition acceptance and extracted spacing as the PicoScope trigger was varied. The stable high-acceptance region begins near 25 mV.

5 Waveform Processing

5.1 Event-by-event baseline

Electronic offset was estimated for each event from pre-trigger samples. A robust procedure was used instead of a plain mean. The median and median absolute deviation were first calculated,

$$\text{MAD} = \text{median}(|x_i - \text{median}(x)|), \quad \sigma_{\text{robust}} = 1.4826 \text{ MAD}. \quad (6)$$

Outlying pre-trigger samples were removed and the baseline was recalculated. The corrected waveform was

$$v(t) = V_{\text{recorded}}(t) - V_{\text{baseline}}. \quad (7)$$

The robust estimate prevents one pickup transient or early pulse from moving the baseline of an entire event.

5.2 Pulse location and integration

The pulse maximum was searched for only within the expected prompt region. Searching the entire record can incorrectly select a late afterpulse or pickup transient. A three-point quadratic interpolation estimated the maximum between ADC samples. The integration gate in [equation \(5\)](#) then followed the measured peak time.

Waveforms were rejected when the baseline region was incomplete, the prompt maximum fell outside the permitted interval, the pulse failed the noise requirement, the integration gate left the recorded window, or the waveform touched the digitizer rail.

5.3 Saturation and quality control

Saturation was checked using waveform overlays and the relationship between height and area. Clipping is indicated when many traces share the same maximum, peak height stops increasing while area continues to change, or samples accumulate at the scope rail. A clipped spectrum cannot be repaired by recording more events; the range, gain, or bias must be changed.

The accepted-event fraction was also plotted against voltage. An abrupt acceptance change can indicate trigger selection rather than a real change in SiPM behavior.

6 Photoelectron Spectrum Analysis

6.1 Historical peak-finding method

The early analyses built a raw histogram and smoothed its bin counts with `scipy.ndimage.gaussian_filter1d`. Candidate maxima were found with `scipy.signal.find_peaks` and refined using a local Gaussian plus linear background fitted with `scipy.optimize.curve_fit`. The refined centers were sorted, and combinations were tested for an approximately even sequence.

For histogram counts $H(x_i)$, Gaussian smoothing has the form

$$S(x) = \frac{\sum_i H(x_i) \exp [-(x - x_i)^2 / (2\sigma_s^2)]}{\sum_i \exp [-(x - x_i)^2 / (2\sigma_s^2)]}. \quad (8)$$

The smoothed curve was used only to locate candidates. Peak centers were refined against the unsmoothed bins. This distinction matters because smoothing does not add information to the data.

6.2 Pedestal-constrained comb model

The current primary method fits several p.e. populations together. Their centers share one spacing,

$$\mu_n = \mu_0 + n\Delta Q. \quad (9)$$

The pedestal location and width are constrained by the independent pedestal acquisition. Population amplitudes and widths can vary. The optimization uses `scipy.optimize.least_squares` with a Poisson-deviance residual for histogram counts. Several starting spacings are tried to test stability.

The multi-population model is compared with a single broad population using the Bayesian information criterion,

$$\text{BIC} = D + k \ln N, \quad (10)$$

where D is the deviance, k the number of model parameters, and N the number of fitted observations used by the implementation. In the lower-bias study, a comb point required $\Delta\text{BIC} \geq 10$ together with stable spacing and separated neighboring populations.

6.3 Why every visible maximum is not a p.e. peak

A p.e. sequence should be approximately periodic. A small first population followed by a larger second population can be physically reasonable because the trigger can suppress small pulses and optical crosstalk can move events into higher populations. However, two nearby maxima inside one expected population can also result from pulse-shape variation, pickup, or over-flexible peak finding. The analysis therefore uses the complete spacing pattern and fit stability rather than accepting every local maximum.

6.4 Spacing and breakdown-voltage uncertainties

The comb-fit covariance gives the statistical uncertainty of ΔQ under the selected model. The spacing points are then fitted against bias with an uncertainty floor so that one unrealistically precise point cannot dominate. If the point scatter exceeds the stated errors, an additional vertical scatter term is included.

The covariance of slope and intercept is propagated through $V_{\text{br}} = -b/m$. The fit is also repeated while removing one voltage at a time. A large movement under this leave-one-out test indicates that the nominal regression uncertainty is not a reliable description of the measurement.

SiPM 1 (Δ): pulse-area spectra

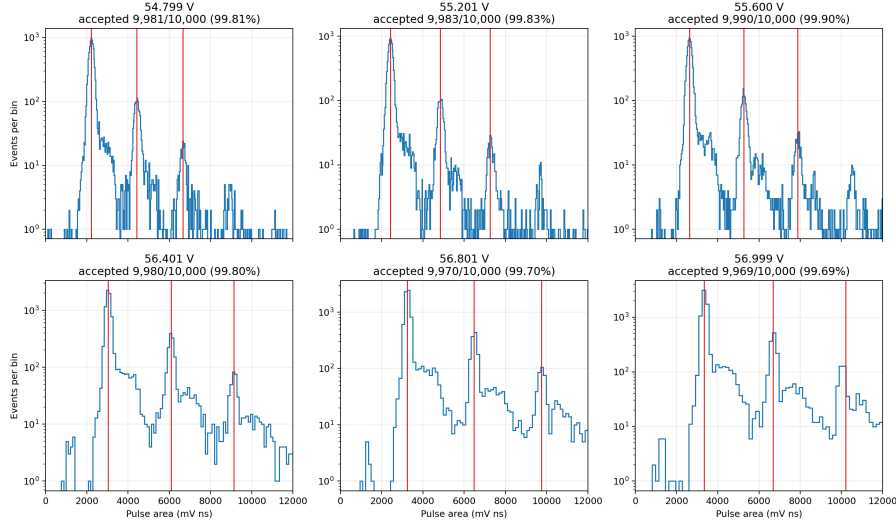


Figure 4: Pulse-area spectra from the SiPM 1 (Δ) across the clean bias scan. Raw histogram structure and the fitted population sequence were inspected at each voltage.

SiPM 2 ($\star$): pulse-area spectra

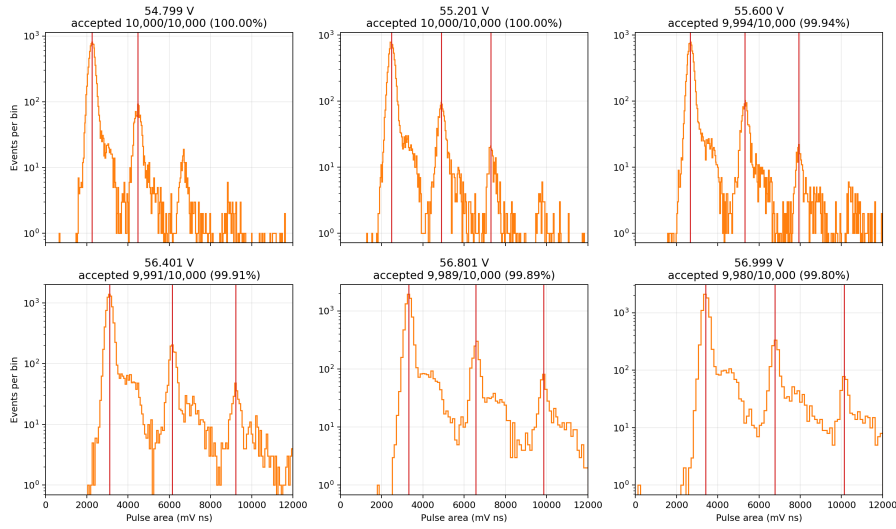


Figure 5: Pulse-area spectra from the SiPM 2 ($\star$) across the clean bias scan.

7 Breakdown-Voltage Results

7.1 Original SiPM pair

The original SIPM1 and SIPM2 were measured in three independent scans. Each main scan used six bias points from approximately 53.4 to 56.9 V and 10,000 waveforms per SiPM per point. The combined internal results at 20 °C were:

Table 2: Breakdown-voltage results for the original SiPM pair.

Device	V_{br} (V)	Area-spacing slope (mV ns/V)	R^2
SIPM1	50.637 ± 0.037	532.5	0.9997
SIPM2	51.988 ± 0.103	555.9	0.9976

The two devices have clearly different operational intercepts on the present bias scale. Applying one common voltage would therefore place them at different nominal overvoltages.

An interleaved scan tested new voltage points against previously frozen lines. The new intercepts, (50.6466 ± 0.0374) V and (52.0662 ± 0.0633) V, were compatible with the frozen values. This gave an important check that the fitted lines were not only reproducing the exact voltage points used to create them.

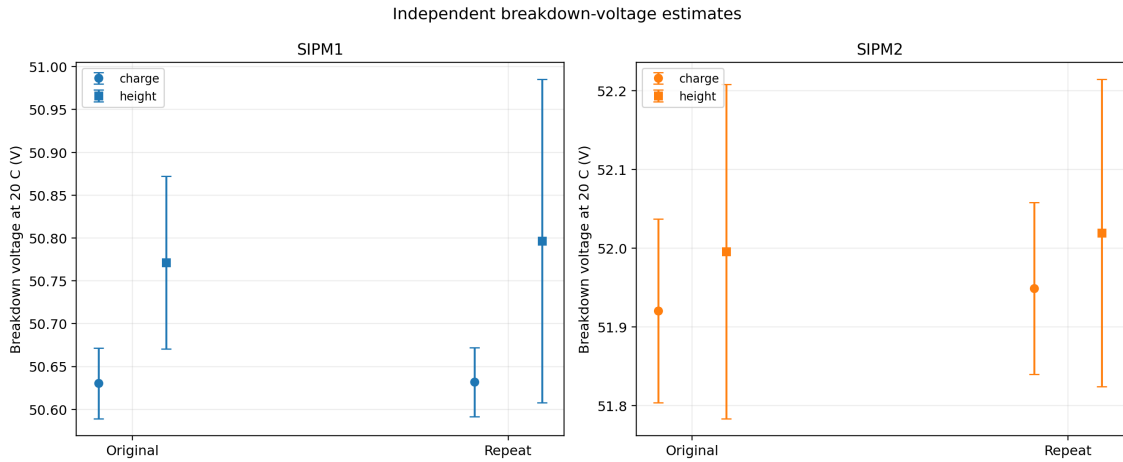


Figure 6: Breakdown-voltage repeatability for the original SiPM pair.

7.2 Lower-bias limit

A separate scan covered 52.0 to 53.4 V in 0.2 V steps. The continuous stable region began at 52.6 V for SIPM1 and 53.2 V for SIPM2. Below those values, neighboring populations overlapped, trigger acceptance decreased, or the fitted spacing depended strongly on starting values.

For SIPM1, adding every apparently resolved low-bias point moved the fitted intercept from 50.630 to 50.800 V and increased the required intrinsic charge scatter from 7.9 to 25.5 mV ns. The added points therefore introduced low-end selection bias rather than useful leverage.

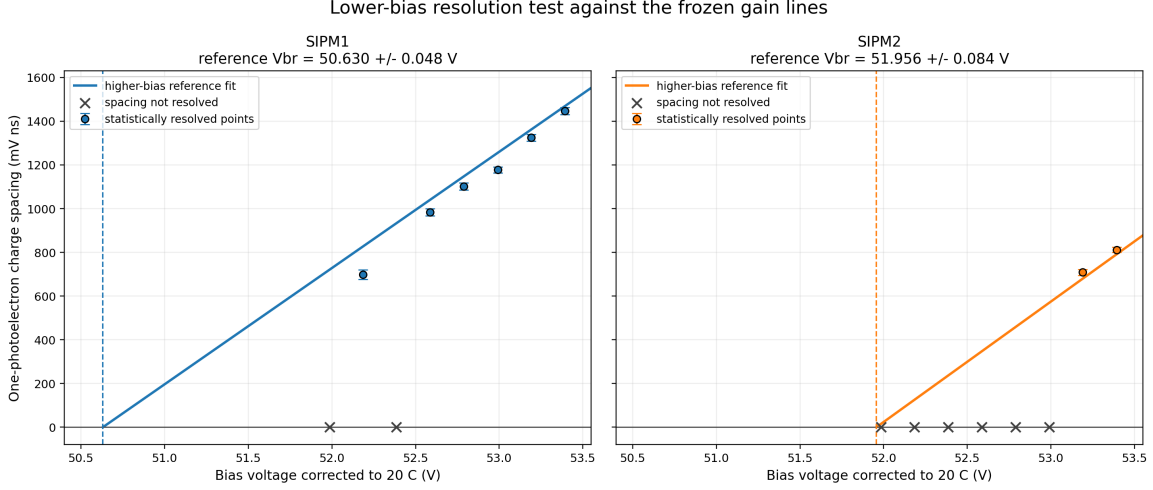


Figure 7: Lower-bias resolution tests. Points close to breakdown were retained only when the multi-population structure remained statistically supported and stable.

7.3 SiPM 1 ($\triangle$) and SiPM 2 ($\star$)

After the CH3 trigger study, both physical SiPMs were measured over seven matched bias points with 10,000 waveforms at each point. The complete scan was then repeated independently. The temperature during the repeat ranged from 20.365 to 20.462 °C.

Table 3: Two clean measurements of SiPM 1 ($\triangle$) and SiPM 2 ($\star$). Uncertainties are internal waveform-analysis uncertainties.

Device	First scan V_{br} (V)	Repeat V_{br} (V)	Weighted V_{br} (V)
SiPM 1 ($\triangle$)	50.534 ± 0.113	50.486 ± 0.121	50.512 ± 0.083
SiPM 2 ($\star$)	50.667 ± 0.097	50.488 ± 0.092	50.574 ± 0.067

The repeat scan accepted 69,860 of 70,000 SiPM 1 ($\triangle$) waveforms and 69,952 of 70,000 SiPM 2 ($\star$) waveforms. The repeat area-spacing slopes were 513.32 ± 12.44 and 515.68 ± 9.40 mV ns V⁻¹, respectively. The mean absolute point-by-point changes in spacing were 0.66% and 0.59%.

The SiPM 2 ($\star$) minus SiPM 1 ($\triangle$) breakdown-voltage difference is

$$\Delta V_{br} = (0.062 \pm 0.106) \text{ V}, \quad (11)$$

which is a 0.58σ difference. The current data therefore do not show a statistically significant difference between these two devices.

7.4 Provisional operating points

A controlled starting point is 3.0 V above the measured breakdown voltage. Near 20.4 °C, this gives approximately

These are working settings on the present bias calibration. They should not be interpreted as traceable terminal voltages until the high side and channel low sides are measured directly with a calibrated high-impedance instrument.

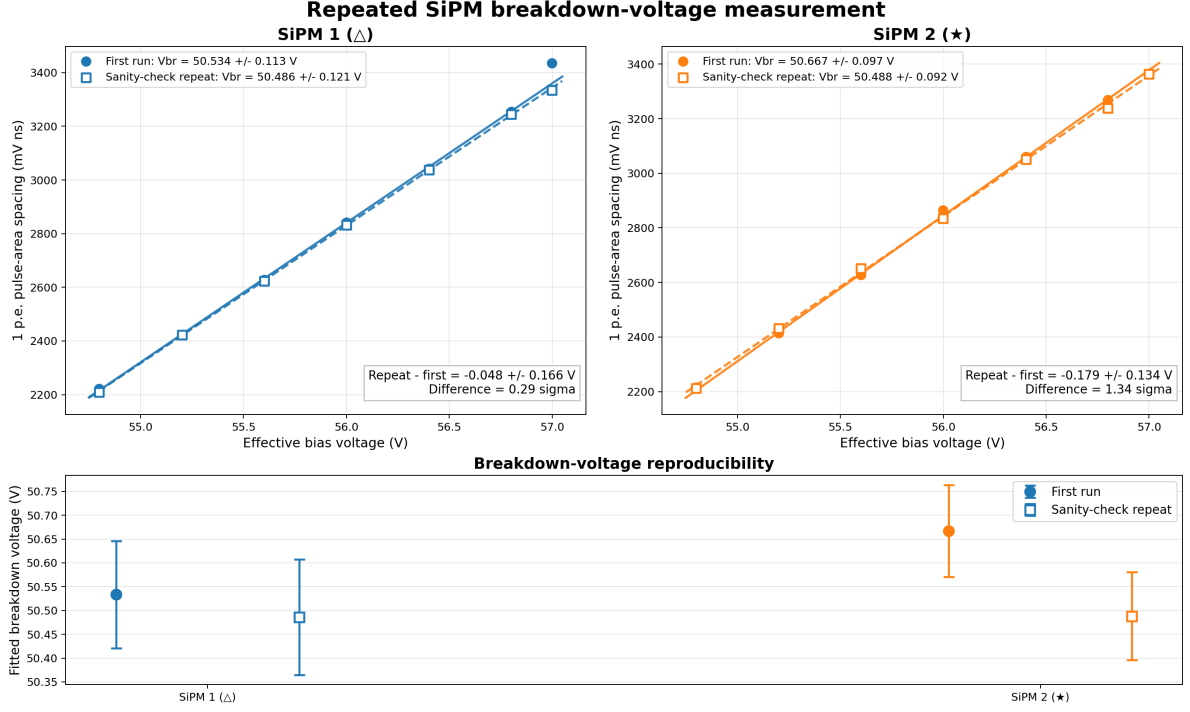


Figure 8: First and repeated breakdown-voltage measurements for SiPM 1 (Δ) and SiPM 2 ($\star$). The channel-switched and repeated points provide a test of physical-device identity and run-to-run stability.

Table 4: Provisional operating points for SiPM 1 (Δ) and SiPM 2 ($\star$).

Device	Weighted V_{br} (V)	Provisional $V_{br} + 3.0$ V setting (V)
SiPM 1 (Δ)	50.512 ± 0.083	53.51
SiPM 2 ($\star$)	50.574 ± 0.067	53.57

8 EPIC-Triggered Zero-Event Study

8.1 Purpose

The breakdown-voltage measurement equalizes nominal overvoltage. It does not test the final response to scintillation light. Two SiPMs at the same overvoltage can still differ in photon-detection efficiency, dark-count rate, correlated noise, fibre coupling, or amplifier response.

The zero-event study uses two EPIC tiles as reference counters. A top and bottom EPIC coincidence selects a particle crossing the central region. The two GSU gLOWCOST tile waveforms are then examined for a pulse in the expected time window.

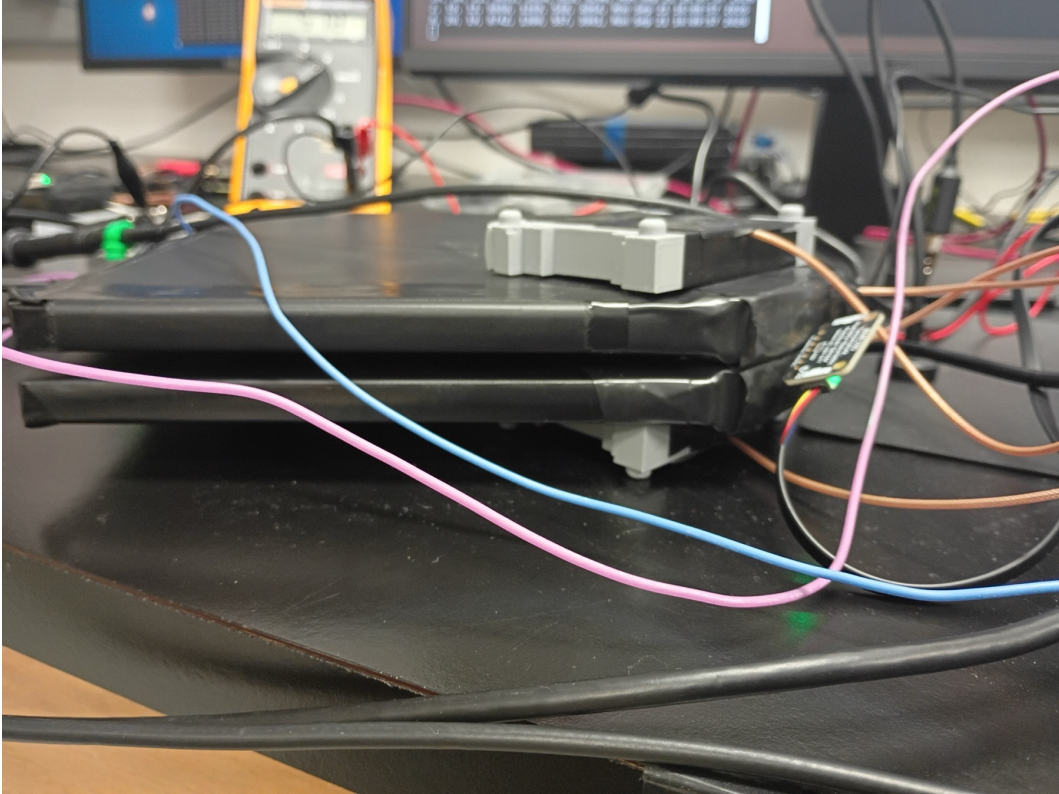


Figure 9: Two GSU gLOWCOST tiles placed between the top and bottom EPIC reference tiles.

The current physical mapping is shown in [table 5](#).

Table 5: Channel mapping for the EPIC-triggered GSU gLOWCOST measurement.

Position	SiPM label	PCB channel	Scope channel
Top EPIC tile	not individually calibrated	CH0	C
Top GSU gLOWCOST tile	SiPM 1 ($\triangle$)	CH2	A
Bottom GSU gLOWCOST tile	SiPM 2 ($\star$)	CH3	B
Bottom EPIC tile	not individually calibrated	CH1	D

8.2 Zero fraction and dark correction

For each EPIC coincidence, the GSU gLOWCOST signal gate is classified as zero or nonzero using a boundary between the pedestal and one-p.e. response. An equal-duration dark-time

window from the same waveform estimates accidental dark pulses and electronic-noise crossings.

Let f_s be the measured zero fraction in the expected signal gate and f_d the zero fraction in the dark-time gate. If the scintillation response and accidental dark activity are independent, an empty signal gate requires both no detected scintillation response and no accidental pulse. Therefore,

$$P_0(\text{scintillation}) = \frac{f_s}{f_d}, \quad \epsilon_{\text{chain}} = 1 - \frac{f_s}{f_d}. \quad (12)$$

The correction prevents an accidental dark pulse from being counted as successful scintillation detection.

If detected primary photoelectrons follow a Poisson distribution with mean μ , then

$$P(0) = e^{-\mu}, \quad \mu = -\ln P(0). \quad (13)$$

For cosmic-ray scintillation, μ should be described as a zero-equivalent occupancy rather than an absolute photon yield. Muons do not all deposit identical energy or follow identical paths, and the optical collection varies with position. A pulsed LED with controlled light is required for a cleaner relative PDE measurement.

8.3 Pilot observations

The first EPIC coincidence attempt used high analogue thresholds and produced only auto-trigger records, showing that the condition was too strict. Later single-channel tests showed that scope D pulses were smaller than scope C pulses. An HV-off control removed the live pulse population, confirming that the observed coincident signals were detector generated rather than a persistent scope artifact.

A 5,000-event development run used the following provisional reference cuts:

- scope C positive excursion of at least 3000 mV;
- scope D positive excursion of at least 120 mV; and
- absolute C–D peak-time difference no greater than 40 ns.

Six events passed these strict cuts in 81.49 s, corresponding to approximately 4.4 events per minute. Four of the six showed excursions above 200 mV in both GSU gLOWCOST channels. This sample demonstrated the geometry and timing logic but was too small for an efficiency result.

8.4 Production acquisition

8.5 Zero-event results

Scintillator pilot: 50 four-channel triggers

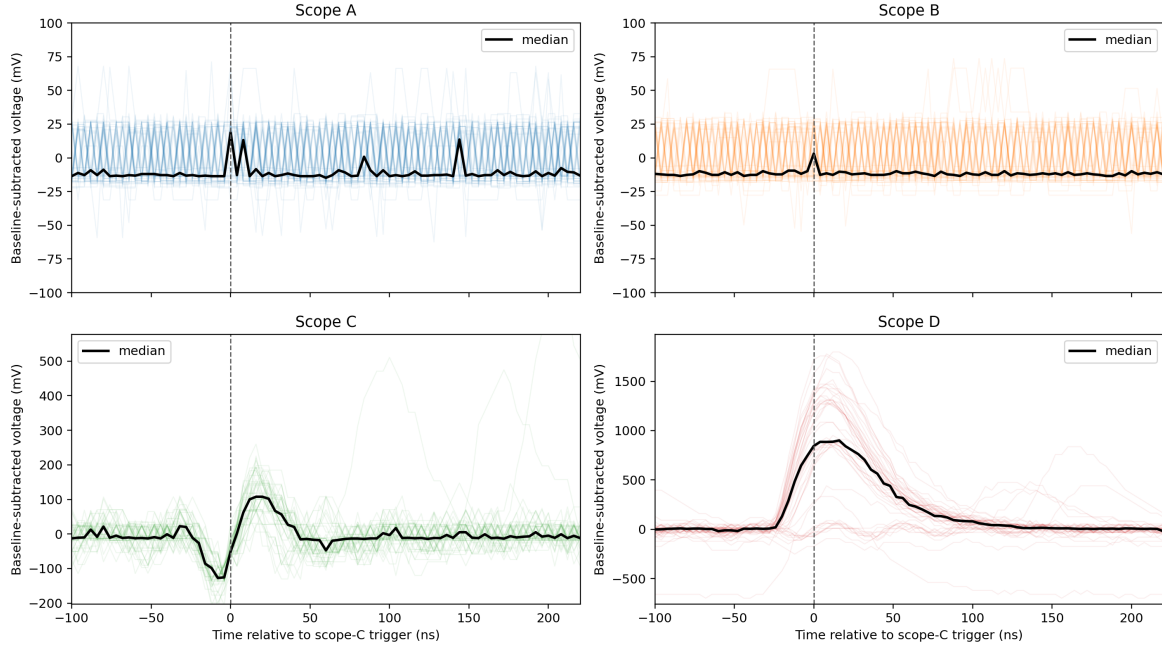


Figure 10: Four-channel waveform pilot for the scintillator zero-event method. The pilot established channel timing and pulse polarity; it was not used as a final efficiency measurement.

9 Discussion

9.1 Temperature

10 Next Measurements

11 Conclusion

Acknowledgements

References

- [1] Hamamatsu Photonics K.K. *MPPC Technical Information*. Hamamatsu Photonics K.K. Accessed September 2026.
- [2] Robert Klanner. Characterisation of sipms. *Nuclear Instruments and Methods in Physics Research Section A*, 926:36–56, 2019.
- [3] Hamamatsu Photonics K.K. *MPPC S13360 Series Datasheet*. Hamamatsu Photonics K.K. Accessed September 2026.

- [4] V. Chmill, E. Garutti, R. Klanner, M. Nitschke, and J. Schwandt. On the characterisation of sipms from pulse-height spectra. *Nuclear Instruments and Methods in Physics Research Section A*, 854:70–81, 2017.

A Software and Reproducibility

The analysis was performed with Python 3.10, NumPy, pandas, SciPy, and Matplotlib. PicoScope software was used for early manual recordings, and PicoSDK was used for automated acquisition. The main analysis functions used were:

- `numpy.trapezoid` for pulse-area integration;
- `scipy.ndimage.gaussian_filter1d` for diagnostic histogram smoothing;
- `scipy.signal.find_peaks` for historical candidate discovery;
- `scipy.optimize.curve_fit` for historical local Gaussian fits;
- `scipy.optimize.least_squares` for the simultaneous p.e. comb; and
- `scipy.optimize.brentq` for the additional-scatter solution.

The code, compact processed data, figures, and chronological laboratory notes are available at <https://github.com/tharinduudu/SiPM-Operating-Point-Study>. Raw waveform campaigns remain on the acquisition computer because their size is not suitable for Git.

B Important Distinction Between Thresholds

Three different thresholds appear in the study:

1. the PicoScope trigger used to decide which waveforms are recorded;
2. the software boundary used to classify a waveform as zero or nonzero; and
3. the hardware comparator threshold used by the operating detector.

They have different purposes and must not be interchanged. The 25 mV value used to correct CH3 waveform acquisition is not a recommendation for the normal detector comparator threshold.

C Summary of Uncertainty Categories

Table 6: Uncertainty categories and their current treatment.

Source	Treatment
P.e.-spacing statistics	Included in each spectrum-fit uncertainty
Slope-intercept covariance	Included in the internal V_{br} uncertainty
Excess point scatter	Added when the observed scatter exceeds point errors
Run-to-run repeatability	Included when compatible independent runs are combined
Absolute high-side calibration	Not yet included
Per-channel DAC calibration	Not yet fully included as a traceable systematic
Temperature coefficient	Provisional family value; individual scan pending
Trigger selection	Tested explicitly; problematic points rejected
Analogue transfer calibration	Not included in absolute charge or gain